CentraleSupélec | université PARIS-SACLAY

Project description — Applied Artificial Intelligence for Health

2026-2027

Transplant or Wait

Prognostic task formulation and clinical evaluation in myelodysplastic syndromes

1- Context:

Keywords: ***clinical decision support, prognostic modelling, model evaluation, deployment***

Myelodysplastic syndromes are clonal haematopoietic disorders characterised by ineffective haematopoiesis, peripheral cytopenias and a variable risk of progression to acute myeloid leukaemia. Clinical trajectories are markedly heterogeneous: a substantial proportion of patients remain stable for several years under surveillance alone, whereas others deteriorate within months. At diagnosis these divergent trajectories are difficult to distinguish from clinical presentation alone.

Therapeutic intensity is decided at diagnosis, in a multidisciplinary meeting, between surveillance and supportive care, hypomethylating agents, and allogeneic haematopoietic stem cell transplantation. Transplantation remains the only potentially curative option and carries substantial treatment-related mortality. The decision is therefore governed by an asymmetric trade-off: over-treatment exposes low-risk patients to procedural mortality, while under-treatment forfeits the only curative option in patients whose disease progresses. The two errors fall on distinct patient populations and cannot be reconciled through a single scalar loss.

Risk stratification currently relies on established prognostic scores. The Revised International Prognostic Scoring System combines cytopenias, marrow blast percentage and cytogenetic risk category [1]; its molecular refinement additionally incorporates somatic mutations and reclassifies a substantial fraction of patients [2]. Any candidate tool is therefore assessed against an existing comparator, and comparative evaluation is a requirement of the project rather than an optional extension.

Reported failures of predictive systems in clinical settings arise predominantly from task formulation rather than from model estimation [3]: misspecified cohorts, prediction times inconsistent with the clinical workflow, and outputs delivered outside the window in which a decision remains actionable. The emphasis of this project falls accordingly on problem specification and evaluation design.

Teams work under a simulated commissioning scenario in which a haematology department requests a deployable tool. A publicly available multi-centre cohort is supplied without prior specification or preprocessing; whether it can support the requested task is itself to be established. Clinical guidance is available from the instructor, acting as the referring haematologist, subject to a fixed question budget per team. Identifying what cannot be inferred from the data is part of the assessed work.

2- Expected result and deliverables:

A prognostic decision-support tool suitable for presentation at the therapeutic decision meeting, with the evidence supporting its intended use. Three deliverables:

- **A functioning system** and the code that produces it, taking information available at the time of diagnosis and returning an output interpretable at the point of decision.
- **A written report** specifying the intended user and use context, the predicted outcome and prediction time, comparative performance against the reference score, and the conditions under which performance degrades.
- **A ten-minute oral defence** before the commissioning department.

Assessment weights problem formulation, the rigour and honesty of the evaluation, and the characterisation of failure modes. Methodological sophistication is not credited in itself: a parsimonious model addressing a well-specified question is preferred to a complex model addressing an ill-posed one.

A conclusion that the available data cannot support the requested tool, accompanied by a defensible narrower specification, constitutes an acceptable and positively assessed outcome.

3- Bibliography:

[1] Greenberg, P. L., et al. (2012). Revised International Prognostic Scoring System for myelodysplastic syndromes. Blood, 120(12), 2454-2465.

[2] Bernard, E., et al. (2022). Molecular International Prognostic Scoring System for myelodysplastic syndromes. NEJM Evidence, 1(7), EVIDoa2200008.

[3] Wiens, J., et al. (2019). Do no harm: a roadmap for responsible machine learning for health care. Nature Medicine, 25, 1337-1340.